

Review of Related Literature and Studies

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This chapter reviews the literature that grounds the proposed benchmarking of multitask optimization (MTO) on Taglish disaster tweets. The relevant work spans three areas: crisis informatics, MTO, and code-switched text processing. No single area establishes whether specialized balancing benefits triage on code-switched text, so the review covers the three areas in sequence. It arranges prior work by computational subconcept, algorithmic paradigm, and empirical bottleneck rather than by geographical origin.

Section 2.1 reviews crisis informatics corpora, joint-modeling evidence for entity, intent, and urgency tasks, cross-event and cross-lingual generalization, and the computational, statistical, and methodological standards the benchmark adopts. Section 2.2 then turns to MTO. It analyzes the gradient-surgery and dynamic loss-weighting families, the optimizer confound, and the controlled comparisons questioning whether specialized balancing outperforms tuned static linear scalarization. Section 2.3 narrows to the target domain, covering code-switching challenges and measurement, multitask learning on code-switched text, multilingual encoders, and Philippine language resources, including local disaster response studies. Section 2.4 consolidates the key studies in the Algorithmic Literature Taxonomy Matrix, which positions them against the proposed research. Section 2.5 closes with a three-step synthesis that derives the research gap and maps the study’s contributions onto it.

Database searches conducted as of 2026 on MTO, crisis informatics, and code-switching identified the sources. These searches drew primarily on peer-reviewed

conference and journal publications indexed in the ACM Digital Library, IEEE Xplore, SpringerLink, ScienceDirect, and the Association for Computational Linguistics (ACL) Anthology. Within this set, the crisis informatics and code-switching sources that established the evaluation domain date mostly from the last five years. The review retained older studies only if they defined a benchmarked algorithm, a candidate encoder, a methodological and statistical protocol, or a foundational corpus or metric.

2.1 Joint Multitask Modeling and Computational Efficiency in Crisis Informatics

Section 2.1 traces the crisis informatics arc from annotated corpora through joint triage models to cross-event generalization. It then sets the computational, statistical, and methodological standards the benchmark adopts.

2.1.1 Evolution of Crisis Informatics Corpora and Models

Both the corpus track and the model track begin with a demonstration that disaster tweets can be annotated and classified at scale. Imran et al. (2016) collected messages during 19 crises between 2013 and 2015 and released annotations for approximately 50,000 of them. The authors also published word embeddings trained on 52 million crisis-related tweets. The annotation schemes incorporated input from formal response agencies such as the United Nations Office for the Coordination of Humanitarian Affairs. The resulting nine categories included “caution and advice,” “displaced people and evacuations,” and “infrastructure and utilities damage.” Naive Bayes, random forest, and support vector machine classifiers reached an area under the curve (AUC) of at least .80 for most categories, versus .50 for random classification. The exception was the smallest class, “missing, trapped, or found people.” The curation pipeline also flagged

out-of-vocabulary terms such as slang, abbreviations, and misspellings, a noise thread that Section 2.3 develops.

Corpus supply then kept growing, but dispersed label schemes left results impossible to compare. Alam, Sajjad, et al. (2021) documented that gap and responded with CrisisBench, consolidating eight human-annotated datasets into two corpora of 166,098 and 141,533 tweets. Curation removed three forms of duplication, including the near-duplicate content that retweets produce, through cosine similarity filtering. Removing those copies kept identical messages out of both training and test splits.

Corpus construction then scaled up under explicit agreement control. Alam, Qazi, et al. (2021) released the Human-Annotated Disaster Incidents Data (HumAID), sampling tweets from a pool of about 24 million messages across 19 disaster events between 2016 and 2019. Crowd workers read 77,196 sampled tweets under three judgments per tweet, and a label was kept when two of the three agreed, a two-thirds (66%) agreement threshold. Combining the events for cross-event experiments triggered a further near-duplicate removal that left 76,466 tweets.

On these standardized benchmarks, fine-tuned encoders displaced the classical classifiers. CrisisBench's consolidated humanitarian task set the pattern: A fine-tuned RoBERTa model reached .872 on the weighted F1-measure (F1), against .829 for the strongest convolutional-network baseline. A fine-tuned BERT (Bidirectional Encoder Representations from Transformers) model trailed RoBERTa by only .012 weighted F1 on the same task. Evaluations on HumAID showed the same ordering on its combined data. The strongest transformer reached .781 weighted F1, whereas the support vector machine, random forest, and fastText classifiers stayed at or below .712. Two cautions qualify these

numbers. Weighted F1 averages per-class scores with class-frequency weights, so majority classes dominate the total; the benchmark adopted it precisely to factor in class imbalance. The two benchmarks differ in corpora, label sets, and splits, so their numbers support separate conclusions rather than one ranking.

Pre-training itself then moved into the crisis domain. Lamsal et al. (2024) trained CrisisTransformers on more than 15 billion tokens drawn from over 30 crisis events and evaluated the models on 18 crisis-specific public datasets. The domain-adapted models outperformed strong baselines across all of those datasets on classification, and the best sentence encoder improved the state of the art by 17.43%. The pre-training run, however, occupied six A100 GPUs for two months, a cost profile that only a cluster supports.

The current frontier couples triage with generative large language models, and it inherits the cluster habit. Yin et al. (2026) instruction fine-tuned a large language model for multilabel disaster classification and reported 63.8% overall accuracy against 4.27% for zero-shot prompted baselines, a nearly fifteenfold margin. LoRA tuning used a 3-GPU A40 node, and full-parameter tuning ran on a 16-GPU A100 node. Zguir et al. (2025) extended triage toward actionable requests and offers through a hierarchical taxonomy and query-specific few-shot learning, evaluating on 300 manually annotated tweets alongside 1,346 synthetic examples. Results of this generation therefore rest on thin evaluation and on hardware beyond a local workstation.

Across both tracks, three absences mark the space the proposed benchmark enters. First, every corpus in the line stopped at document-level labels; none paired token-level entity spans with concurrent sequence-level intent and urgency labels. Second, the systems behind the strongest results trained on clusters or called cloud services. Third, the

corpus benchmarks reported accuracy and F1 but no inference latency or memory. These three absences, not any single result, define the gap, and Section 2.5 returns to them.

2.1.2 Joint Modeling of Token-Level and Sequence-Level Triage Tasks

The joint-modeling question predates crisis informatics by two decades. Caruana (1997) defined multitask learning as inductive transfer that improves generalization by using the domain information in the training signals of related tasks. Models learn tasks in parallel over a shared representation, so what the model learns for one task helps the others. Baxter (2000) supplied the theory: A hypothesis space that performs well on sufficiently many related tasks will also perform well on novel tasks from the same environment.

The nearest structural precedent comes from spoken-language understanding. Weld et al. (2022) surveyed the joint models behind conversational agents and chatbots. The standard approach trains models for intent detection over the whole utterance together with slot filling over its individual tokens. This pairing exhibits the structural configuration that the proposed triage formulation requires: one token-level task alongside sequence-level tasks.

Prior crisis informatics studies applied joint modeling only partially. Wang et al. (2021) fine-tuned one BERT model on the TREC Incident Streams (TREC-IS) crisis corpus to predict information types and priority levels together. The loss weight remained fixed at .50, weighting both task objectives equally. Seeberger and Riedhammer (2022) combined entity-masked language modeling with hierarchical multilabel classification on the same TREC-IS corpus under its 2020B split. The combined design gained up to 10 percentage points of F1 on actionable information types, and entity masking reduced

overfitting to in-domain events.

Prior crisis studies have not evaluated this concurrent pairing. No reviewed study evaluated token-level entity extraction together with intent and urgency classification on the same disaster tweets, and none examined the optimization strategies such architectures demand. Whether specialized balancing resolves that shared loss is the question Section 2.2 takes up.

2.1.3 Cross-Event and Cross-Lingual Generalization

Evaluation practice moved toward held-out events in the same years. Ray Chowdhury et al. (2020) trained multilingual BERT with Manifold Mixup and tested on four datasets, three of which came from disasters absent from the training data. On the English-only evaluation, the model reached an F1 of 79.36% on the Philippines Flood event. Zero-shot transfers scored 81.33% on French, 75.44% on Italian, and 85.26% on Spanish. Per-class evaluation on the multilingual mixed set showed the lowest score on the caution and advice class, at 70.3% F1.

Cross-lingual evaluations also quantified the performance drop incurred during low-resource transfer. Sarioglu Kayi et al. (2020) trained urgency classifiers on English crisis tweets and released evaluation sets for Sinhala and Odia. The ensemble classifier scored 76.5% F1 on English, 63.5% on Sinhala, and 62.6% on Odia when transferred without target-language training. The authors traced part of the noise to annotation because general status updates and critical help requests both carried the urgent label.

Recent work treated generalization failure itself as a modeling problem. Seeberger et al. (2025) showed that crisis classifiers relied on spurious event-specific cues such as

locations and hashtags. The authors proposed a causal debiasing framework with domain-specific experts. Across three disaster classification tasks under temporal splits to unseen events, the approach outperformed the next-best baseline by 1.1 to 1.9 percentage points and significantly improved the BERT baseline. This line of evidence justifies the leave-one-event-out (LOEO) evaluation of the proposed benchmark, which holds out one typhoon at a time.

2.1.4 Complexity Analysis, Statistical Rigor, and Design Science Methodology

The benchmark’s complexity axis rests on a structural property of shared encoders. One forward pass over the encoder serves all three triage tasks, whereas three single-task systems pay the encoder cost three times per tweet. Inference latency and memory therefore become measurable benchmark axes in their own right. The MTO landscape is also consolidated enough to compare: LibMTL packages 13 optimization strategies and eight architectures behind one reproducible PyTorch interface ([Lin & Zhang, 2023](#)).

Single reported scores carry a variance risk that single runs hide. Reimers and Gurevych ([2017](#)) retrained sequence-tagging networks with identical hyperparameters but different random seeds. Maximum score differences reached 2.59 F1 points on the Conference on Computational Natural Language Learning (CoNLL) 2003 named entity recognition benchmark. The gap reached 8.23 F1 points on the Automatic Content Extraction (ACE) 2005 Entities task. Selecting a run by its development score was also unreliable, with Spearman’s rank correlation between development and test scores at only $\rho = .23$ in the reported case. The proposed benchmark therefore trains every configuration under multiple seeds and treats single scores as insufficient evidence.

Comparison across many systems and datasets adds its own statistical constraints. Demšar (2006) showed that parametric tests rest on assumptions that comparisons across multiple datasets violate. The recommended alternative is the Friedman test with corresponding post hoc procedures, using the Nemenyi procedure for all-pairwise comparisons, together with the Iman–Davenport correction for the conservative Friedman statistic. The proposed benchmark adopts this nonparametric protocol for its method comparisons.

The last standard concerns the study’s form. Peffers et al. (2007) formalized design science research methodology as a process model of six activities in a nominal sequence, a framework for research that creates and evaluates artifacts. The proposed study follows that framing: The benchmark corpus and the prototype triage service are its design artifacts. No reviewed crisis study combines latency and memory measurement, multi-seed evaluation, nonparametric comparison, and artifact framing in one benchmark, which is the standards gap that Section 2.5 assembles.

References

- Alam, F., Qazi, U., Imran, M., & Ofli, F. (2021). HumAID: Human-annotated disaster incidents data from Twitter with deep learning benchmarks. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2021)*.
<https://doi.org/10.1609/icwsml.v15i1.18116>
- Alam, F., Sajjad, H., Imran, M., & Ofli, F. (2021). CrisisBench: Benchmarking crisis-related social media datasets for humanitarian information processing. *Proceedings of the International AAAI Conference on Web and Social Media*

(ICWSM 2021). <https://doi.org/10.1609/icwsm.v15i1.18115>

Baxter, J. (2000). A model of inductive bias learning. *Journal of Artificial Intelligence Research*. <https://doi.org/10.1613/jair.731>

Caruana, R. (1997). Multitask learning. *Machine Learning*.
<https://doi.org/10.1023/A:1007379606734>

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*. <https://www.jmlr.org/papers/v7/demsar06a.html>

Imran, M., Mitra, P., & Castillo, C. (2016). Twitter as a lifeline: Human-annotated Twitter corpora for NLP of crisis-related messages. *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC 2016)*.
<https://doi.org/10.63317/43bfut6ipc3j>

Lamsal, R., Read, M. R., & Karunasekera, S. (2024). CrisisTransformers: Pre-trained language models and sentence encoders for crisis-related social media texts. *Knowledge-Based Systems*. <https://doi.org/10.1016/j.knosys.2024.111916>

Lin, B., & Zhang, Y. (2023). LibMTL: A python library for deep multi-task learning. *Journal of Machine Learning Research*.
<https://www.jmlr.org/papers/v24/22-0347.html>

Peffer, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*. <https://doi.org/10.2753/MIS0742-1222240302>

Ray Chowdhury, J., Caragea, C., & Caragea, D. (2020). Cross-lingual disaster-related multi-label tweet classification with Manifold Mixup. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics: Student*

Research Workshop. <https://aclanthology.org/2020.acl-srw.39>

Reimers, N., & Gurevych, I. (2017). Reporting score distributions makes a difference:

Performance study of LSTM-networks for sequence tagging. *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing.*

<https://doi.org/10.18653/v1/D17-1035>

Sarioglu Kayi, E., Nan, L., Qu, B., Diab, M., & McKeown, K. (2020). Detecting urgency status of crisis tweets: A transfer learning approach for low resource languages.

Proceedings of the 28th International Conference on Computational Linguistics.

<https://aclanthology.org/2020.coling-main.414>

Seeberger, P., Freisinger, S., Bocklet, T., & Riedhammer, K. (2025). Generalizing to

unseen disaster events: A causal view. *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics.*

<https://doi.org/10.18653/v1/2025.findings-ijcnlp.2>

Seeberger, P., & Riedhammer, K. (2022). Enhancing crisis-related tweet classification

with entity-masked language modeling and multi-task learning. *Proceedings of the Second Workshop on NLP for Positive Impact (NLP4PI).*

<https://doi.org/10.18653/v1/2022.nlp4pi-1.9>

Wang, C., Nulty, P., & Lillis, D. (2021). Transformer-based multi-task learning for

disaster tweet categorisation. *Proceedings of the 18th International Conference on Information Systems for Crisis Response and Management (ISCRAM 2021).*

<https://arxiv.org/abs/2110.08010>

Weld, H., Huang, X., Long, S., Poon, J., & Han, S. C. (2022). A survey of joint intent

detection and slot-filling models in natural language understanding. *ACM Computing Surveys*. <https://doi.org/10.1145/3547138>

Yin, K., Li, B., Liu, C., Mostafavi, A., & Hu, X. (2026). CrisisSense-LLM: Instruction fine-tuned large language model for multi-label social media text classification in disaster informatics. *Advanced Engineering Informatics*.
<https://doi.org/10.1016/j.aei.2026.104836>

Zguir, A. E. F., Ofli, F., & Imran, M. (2025). Detecting actionable requests and offers on social media during crises using LLMs. *Proceedings of the International ISCRAM Conference (ISCRAM 2025)*. <https://doi.org/10.59297/r9q00255>